

Problem

A balanced three-phase system has a distribution wire with impedance $2 + j6 \Omega$ per phase. The system supplies two three-phase loads that are connected in parallel. The first is a balanced wye-connected load that absorbs 400 kVA at a power factor of 0.8 lagging. The second load is a balanced delta-connected load with impedance of $10 + j8 \Omega$ per phase. If the magnitude of the line voltage at the loads is 2400 V rms, calculate the magnitude of the line voltage at the source and the total complex power supplied to the two loads.

Step-by-step solution

Step 1 of 2

$$\overline{S}_1 = 400(0.8 + j0.6) = 320 + j240 \text{ KVA}$$

$$\overline{S}_2 = \frac{3V_p^2}{Z_p^*}$$

For the delta connected load $V_L = V_p$

$$\overline{S}_2 = 3 \times \frac{(2400)^2}{10 - j8} = 1053.7 + j842.93 \text{ KVA}$$

$$\overline{S} = \overline{S}_1 + \overline{S}_2 = 1.3737 + j1.0829 \text{ MVA}$$

Let $I = I_1 + I_2$ be the total line current for I1,

$$S_1 = 3V_p I_1^*, \quad V_p = \frac{V_L}{\sqrt{3}}$$

$$I_1^* = \frac{S_1}{\sqrt{3}V_L} = \frac{(320 + j240)}{\sqrt{3}(2400)}$$

$$\Rightarrow I_1 = 76.98 - j57.735$$

Step 2 of 2

For I2, Convert the load to WYE.

$$I_2 = \frac{I_p \sqrt{3}}{-30^\circ} = \frac{2400}{10 + j8} j3 \angle -30^\circ = 273.1 - j289.76$$

$$I = I_1 + I_2 = 350 - j347.5$$

$$V_s = V_L + V_{line} = 2400 + I(3 + j6)$$

$$= 5.185 + j(1.405) \text{ KV}$$

$$\Rightarrow |V_s| = 5.372 \text{ KV}$$